

Program : Diploma in Polymer Technology	
Course Code : 6092D	Course Title: Process Control And Measuring Devices
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To familiarize characteristics of measuring instruments
- To understand temperature, pressure, flow and level measuring instruments
- To generalize various controllers and hydraulic devices used in manufacturing industries

Course Prerequisites:

Topic	Course code	Course name	Semester
Characteristics of instruments Temperature, pressure, flow and level measurements-principle		Applied Physics I&II	1 & 2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Explain the general characteristics of measuring instruments and temperature measuring instruments	16	Understanding
CO2	Illustrate Pressure measuring instruments	14	Understanding
CO3	Explain Flow and level measuring instruments.	16	Understanding
CO4	Describe Controllers, Hydraulic systems and Hydraulic pumps applicable to polymer processing techniques	12	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3			2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the general characteristics of measuring instruments and temperature measuring instruments		
M1.01	List the static and dynamic characteristics of measuring instruments, its definitions and Temperature scales	4	Remembering
M1.02	Explain the principle and working of mercury in glass thermometer and Bimetallic thermometer.	4	Understanding
M1.03	Explain the principle and working of resistance thermometer and thermocouples	4	Understanding
M1.04	Explain the principle and working of optical and radiation pyrometer	4	Understanding
Contents: General characteristics of Instruments - static and dynamic characteristics of Measuring instruments-accuracy, sensitivity, precision, repeatability, reproducibility and fidelity. Temperature scales like Celsius, Fahrenheit, Kelvin and Rankine Temperature measuring devices - mercury in glass thermometer, Bimetallic thermometer, Resistance thermometer and wheat Stonebridge circuit. See beck effect, thermo electric laws and the working of thermo couple in potentiometer circuit. Thermo couples- metal combinations and their ranges, Pyrometers - principle and working of optical and radiation pyrometer			
CO2	Illustrate Pressure measuring instruments		
M2.01	Define the Pressure scales – gauge pressure, absolute pressure and vacuum scales	2	Remembering
M2.02	Explain the construction and features of ‘U’ tube manometer, well type manometer and micro manometer.	3	Understanding

M2.03	Explain the working principle and construction of C-type bourdon tube, Diaphragms and Bellows.	4	Understanding
M2.04	Summarize the pressure measurement by LVDT, strain gauges, resistance type transducers and piezoelectric transducers.	5	Applying
	Series Test I	1	
Content: Pressure scales - gauge pressure, absolute pressure and vacuum scales. Construction and features of 'U' tube manometer, well type manometer and micro manometer. Construction and working C bourdon tube type bellows and diaphragms. Transducers in pressure measurement - LVDT, strain gauges, resistance type transducers and piezoelectric transducers.			
CO3	Explain Flow and level measuring instruments.		
M3.01	List the direct and indirect level measuring instruments	2	Remembering
M3.02	Explain the principle and working of Hook type level indicator, sight glass, Float type level indicator and Air purge system (bubbler type)	4	Understanding
M3.03	State rate of flow, total flow and Bernoulli's theorem	2	Remembering
M3.04	Explain the working principle of and construction of venturi meter and Orifice meter, variable area meter, turbine and lobed impeller flow meter	8	Understanding
Contents: Direct and indirect means of level measurement - Hook type, sight glass, float actuated, and bubbler type. Rate of flow and total flow, Bernoulli's theorem, derivation of Bernoulli's theorem, application of Bernoulli's - Working of venturimeter and orifice meter. Variable area flow meters-Rotometer.Pulse generating turbine meter, lobed type of flow meter.			
CO4	Describe Controllers, Hydraulic systems and Hydraulic pumps applicable to processing techniques		
M4.01	Differentiate between open loop and closed loop control systems with examples	1	Remembering
M4.02	Explain the Pneumatic Controllers and Electronic controllers, ON-OFF control actions. Proportional derivative and integral control actions and its combinations.	4	Understanding
M4.03	Describe the temperature and pressure control systems in polymer processing techniques	3	Understanding

M4.04	Explain the working principle and working of Hydraulic devices. And pumping devices used in industries	4	Understanding
	Series Test – II	1	
Contents: Open loop and closed loop control actions. Control actions – ON - OFF, proportional, integral, derivative and PID with respect to advantage and limitations of each control action. Principle and application of pneumatic controls, electronic controls with any two examples. Hydraulic systems - hydraulic press, derivation of efficiency, hydraulic crane, hydraulic lift, suspended hydraulic lift, hydraulic coupling. Pumping devices - working of hydraulic ram, jet pump, air lift pump, rotary pumps, external gear pump, internal gear pump, Lobe pump and vane pump centrifugal pump- types of casings and piping systems, reciprocating pump, comparison of pumps			

Text / Reference:

T/R	Book Title/Author
R1	A Course in Mechanical measurement and Instrumentation- A.K. Sawhney (Publisher: Dhanpat Rai, New Delhi Edition: 12
R2	Industrial Instrumentation -Donald P Eckman(Publisher Wiley 1951)
R3	Instrumentation -Franklyn.W.Kirk& Nicholas R.Rimboi (Publisher – American Technical Society 1962)
R4	Textbook of Hydraulics, Fluid Mechanics and Hydraulic Machines - R.S.Khurmi (Publisher- S. Chand Limited, 1987)

Online Resource:

Sl.No	Website Link
1	www.shortcoursesportal.com
2	www.controlandinstrumentation.com
3	www.taylorfrancis.com